

Weak lensing of the Lyman- α forest and its cross-correlation with CMB lensing: literature review and the sample-variance-limited estimator

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Abstract

These notes accompany an attempt to detect gravitational lensing of the DESI DR1 Lyman- α forest through its cross-correlation with CMB lensing maps from ACT and Planck. Section 1 reviews the (small) literature on lensing of the forest and the adjacent literature on forest–CMB-lensing correlations and on lensing reconstruction from three-dimensional source fields. Section 2 derives, in the flat-sky, single-source-plane, zero-noise (sample-variance) limit, the optimal quadratic estimator for the convergence from a set of forest pixels, the cubic estimator $W_{ijk}\delta_F^i\delta_F^j\delta_{F\kappa_{\text{CMB}}}^k$ for the cross-spectrum $C_L^{\kappa_{\text{Ly}\alpha}\kappa_{\text{CMB}}}$, its variance, the dense-sightline continuum limit with a closed-form reconstruction noise $N_\kappa(L)$, and a bias-hardened variant that projects out the response of the forest power spectrum to long-wavelength modes. Section 3 evaluates $N_\kappa(L)$ and the cross-correlation signal-to-noise for a fiducial forest power spectrum and DESI-like sightline densities. The central qualitative result is that for sparse sightlines most of the naive sample-variance information comes from an isotropic dilation of the transverse correlation function that is degenerate with a local modulation of the forest power spectrum amplitude, and that the robust (shear-like) information is set by the transverse rather than the line-of-sight resolution.

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1 Literature review

1.1 Lensing of the forest

Only one group has published on reconstructing the lensing potential from the forest itself. Croft et al. [1] proposed the idea: the three-dimensional forest is treated as a stack of source screens at $z \simeq 2\text{--}3$ and a configuration-space quadratic estimator, adapted from 21 cm lensing work, is applied to simulated spectra on a $15^\circ \times 15^\circ$ field with a sightline every 1.8 arcmin (which they call the extreme high end of achievable densities). They recover the degree-scale morphology of the potential and give no per-survey signal-to-noise. Metcalf et al. [2] supplied Gaussian noise forecasts. The noise per potential mode scales as $\sigma_\tau^2/[j_{\max}(l_{\max} - l_{\min})^2 l_{\min}^2]$, where σ_τ is the fractional per-pixel error and $j_{\max}$ the number of radial modes. Their cumulative signal-to-noise on the convergence power spectrum for DESI (770k spectra, 14 000 deg^2 , 8.1 arcmin mean separation, per-pixel S/N 3.3) is 7 for $\Delta z = 0.1$ source bins and 29 for $\Delta z = 0.5$ bins, corresponding to a 1.5 per cent measurement of the amplitude of mass fluctuations at $z \simeq 2.5$; eBOSS gives 4–16 and an MSE-like survey 50–160. Metcalf et al. [3] replaced the harmonic estimator by a real-space minimum-variance pixel-pair estimator built on the full pixel covariance, with the potential expanded in Legendre polynomials over a patch. This handles irregular sightline positions, holes and inhomogeneous noise. On Gaussian mocks they find that high-fidelity maps require $\gtrsim 0.5$ sources per arcmin^2 and signal-dominated pixels; a density of 200 deg^{-2} fails. Shaw et al. [4] applied that estimator to hydrodynamic simulations with ray-traced potentials: forest non-Gaussianity reduces the signal-to-noise by a factor of about 2.7 in the noise-free case and 1.5 for unit signal-to-noise spectra, and the recovered potential is biased by 5–25 per cent, which they correct by Gaussianizing the flux field. Shaw et al. [5] forecast DESI directly: 50 quasars per deg^2 , 700k spectra, pixel noise $\sigma_\delta = 0.3$ after rebinning to 2.8 Å and discarding the noisiest quarter, and detection defined as the correlation of the reconstructed potential with the foreground galaxy distribution. The full-survey significance is $S/N \simeq 3$ (two-pixel estimator), 4 (four-pixel) and 9 without pixel noise; 200–2000 deg^{-2} surveys on the same area reach 16–27. They state explicitly that they do not consider cross-correlation with CMB lensing.

No detection of forest lensing has been claimed, no analysis of eBOSS or DESI data exists, and no paper considers the cross-correlation of a forest-lensing reconstruction with CMB lensing. The last point is the gap this project targets: ACT DR6 [6, 7] and Planck PR4 [8] lensing maps cover the full DESI DR1 forest footprint of more than 9500 deg^2 with 428 403 valid Ly α -region forests [9], and a cross-correlation is immune to the additive Gaussian ($N^{(0)}$ -type) bias that dominates an auto-spectrum measurement.

1.2 Forest–CMB-lensing correlations that are not lensing of the forest

A separate line of work correlates the forest flux, or its local one-dimensional power, with κ_{CMB} . Vallinotto et al. [10, 11] forecast the two-point $\langle \delta_F \kappa_{\text{CMB}} \rangle$ and the $\langle \delta_F^2 \kappa_{\text{CMB}} \rangle$ correlations; Doux et al. [12] detected the latter at 5σ with BOSS DR12 and Planck 2015; Chiang and Slosar [13] modelled it as the squeezed gravitational bispectrum plus a term from mean-flux misestimation on finite skewers; Karaçaylı et al. [14] detected it at 4.8σ with DESI Y1 forests and Planck PR3; La Posta and Schaan [15] forecast $S/N \simeq 10, 15$ and 20 for DESI with ACT, Simons Observatory and CMB-S4. Physically these measure the response of the small-scale forest power to long-wavelength modes, quantified with separate-universe simulations by Chiang et al. [16]. This is important here for a negative reason: as shown in Section 2.3, the same response mimics a lensing convergence in a quadratic estimator, so the detected $P_F \times \kappa_{\text{CMB}}$ bispectrum is the leading contaminant of the measurement we want to make. Loverde et al. [17] discuss the related but distinct effect of quasar magnification bias on forest statistics, at the 0.1–1 per cent level.

1.3 Lensing reconstruction from three-dimensional fields

The formalism below follows the CMB quadratic estimator of Hu [18], Hu and Okamoto [19] generalized to a three-dimensional source by Zahn and Zaldarriaga [20], who showed that each line-of-sight Fourier mode acts as an independent screen and that the screens combine with inverse-variance weights. Lu and Pen [21] pointed out that source non-Gaussianity sets a noise floor even without instrumental noise. Pourtsidou and Metcalf [22, 23] developed the 21 cm intensity-mapping version at $z \simeq 2\text{--}3$ including discreteness noise. Foreman et al. [24] computed the bias to the three-dimensional estimator from the matter bispectrum, found it to cost a factor of a few in signal-to-noise for 21 cm surveys, and introduced a bias-hardened estimator; Schaan et al. [25], Schaan and Ferraro [26] generalized the estimator to weakly non-Gaussian sources and introduced the shear-only estimator that is immune to isotropic modulations of the source power. Bias hardening in the CMB context is due to Namikawa et al. [27]; see also Sailer et al. [28], Maniyar et al. [29]. Buncher et al. [30] recently applied the standard and shear-only estimators to projected galaxy clustering as the source, the closest analogue of a sparse-tracer reconstruction. A fuller annotated list is kept in `report/literature_notes.md`.

2 Formalism in the sample-variance limit

2.1 Set-up and conventions

We work in the flat-sky, distant-observer approximation. A point in the forest volume is labelled by an angular position $\boldsymbol{\theta}$ (radians) and a comoving line-of-sight distance χ ; the effective comoving distance to the forest is χ_* ($\chi_* \simeq 3956 h^{-1}\text{Mpc}$ at $z = 2.4$ for a Planck-2018 cosmology). The flux contrast $\delta_F(\boldsymbol{\theta}, \chi)$ is treated as a statistically homogeneous Gaussian random field with the anisotropic (redshift-space) power spectrum $P_F(k_{\parallel}, k_{\perp})$. Its Fourier transform in the mixed angular/radial variables is

$$\delta(\boldsymbol{l}, k_{\parallel}) = \int d^2\theta d\chi \delta_F(\boldsymbol{\theta}, \chi) e^{-i(\boldsymbol{l}\cdot\boldsymbol{\theta} + k_{\parallel}\chi)}, \quad \langle \delta(\boldsymbol{l}, k_{\parallel}) \delta(\boldsymbol{l}', k'_{\parallel}) \rangle = (2\pi)^3 \delta_{\text{D}}^2(\boldsymbol{l} + \boldsymbol{l}') \delta_{\text{D}}(k_{\parallel} + k'_{\parallel}) \mathcal{P}(\boldsymbol{l}, k_{\parallel}), \quad (1)$$

with $\mathcal{P}(\boldsymbol{l}, k_{\parallel}) = P_F(k_{\parallel}, k_{\perp} = l/\chi_*)/\chi_*^2$. Reality of δ_F implies $\delta(-\boldsymbol{l}, -k_{\parallel}) = \delta^*(\boldsymbol{l}, k_{\parallel})$. For a survey of solid angle Ω and radial depth D the coincident delta functions are $(2\pi)^2 \delta_{\text{D}}^2(0) = \Omega$ and $2\pi \delta_{\text{D}}(0) = D$.

Lensing by a single effective source plane at χ_s is described by a projected potential $\phi(\boldsymbol{\theta})$ with deflection $\boldsymbol{\alpha} = \nabla\phi$, convergence $\kappa = -\nabla^2\phi/2$ (so $\kappa(\boldsymbol{L}) = L^2\phi(\boldsymbol{L})/2$) and shear $\gamma_1 = -(\partial_1^2 - \partial_2^2)\phi/2$, $\gamma_2 = -\partial_1\partial_2\phi$, exactly the convention of Hu and Okamoto [19]. The observed forest field is

$$\tilde{\delta}(\boldsymbol{\theta}, \chi) = \delta_F(\boldsymbol{\theta} + \nabla\phi(\boldsymbol{\theta}), \chi). \quad (2)$$

Because the source plane is common to all χ , the displacement is independent of the radial coordinate; this is the single approximation that makes the problem separable in $k_{\parallel}$. In reality the lensing kernel changes by a few per cent across a forest, which we defer to later work.

2.2 Mode coupling induced by lensing

Expanding (2) to first order, $\tilde{\delta} = \delta_F + \nabla\phi \cdot \nabla_{\theta}\delta_F$, and Fourier transforming,

$$\tilde{\delta}(\mathbf{l}, k_{\parallel}) = \delta(\mathbf{l}, k_{\parallel}) - \int \frac{d^2 l_1}{(2\pi)^2} [\mathbf{l}_1 \cdot (\mathbf{l} - \mathbf{l}_1)] \phi(\mathbf{l} - \mathbf{l}_1) \delta(\mathbf{l}_1, k_{\parallel}). \quad (3)$$

For a fixed realization of ϕ the two-point function of the observed field acquires an off-diagonal part. With $\mathbf{L} = \mathbf{l} + \mathbf{l}'$ and $\mathbf{L} \neq 0$,

$$\langle \tilde{\delta}(\mathbf{l}, k_{\parallel}) \tilde{\delta}(\mathbf{l}', k'_{\parallel}) \rangle_{\phi} = 2\pi\delta_D(k_{\parallel} + k'_{\parallel}) \left[(2\pi)^2 \delta_D^2(\mathbf{L}) \mathcal{P}(l, k_{\parallel}) + f(\mathbf{l}, \mathbf{l}'; k_{\parallel}) \phi(\mathbf{L}) \right], \quad f(\mathbf{l}, \mathbf{l}'; k_{\parallel}) = \mathbf{L} \cdot \mathbf{l} \mathcal{P}(l, k_{\parallel}) + \mathbf{L} \cdot \mathbf{l}' \mathcal{P}(l', k'_{\parallel}) \quad (4)$$

It is convenient to express the response in terms of the convergence, $f_{\kappa} \equiv 2f/L^2$, so that the off-diagonal term reads $f_{\kappa} \kappa(\mathbf{L})$. Equation (4) is diagonal in $k_{\parallel}$: every radial Fourier mode is an independently lensed two-dimensional screen with power spectrum $\mathcal{P}(l, k_{\parallel})$, all sharing the same ϕ [20]. Pairs with $k'_{\parallel} \neq -k_{\parallel}$ carry no lensing information and, for a Gaussian field, no covariance either, so an optimal estimator never uses them.

The pair-displacement picture. The same result follows from the real-space statement in the task description: a pair of pixels $i = (a, p)$ and $j = (b, q)$ on sightlines a, b at radial positions χ_p, χ_q has covariance

$$C_{ij}(\phi) = \xi(\boldsymbol{\theta}_{ab} + \boldsymbol{\alpha}_a - \boldsymbol{\alpha}_b, \chi_p - \chi_q) \simeq \xi(\boldsymbol{\theta}_{ab}, \Delta\chi_{pq}) + \nabla_{\theta}\xi(\boldsymbol{\theta}_{ab}, \Delta\chi_{pq}) \cdot [\nabla\phi(\boldsymbol{\theta}_a) - \nabla\phi(\boldsymbol{\theta}_b)], \quad (5)$$

where ξ is the correlation function of δ_F in angular-radial coordinates and $\boldsymbol{\theta}_{ab} = \boldsymbol{\theta}_a - \boldsymbol{\theta}_b$. Only the relative deflection of the two members of a pair matters, so a pair measures the local tidal field of ϕ rather than the deflection itself. Fourier transforming (5) over both positions reproduces (4); this was verified symbolically for a periodic one-dimensional toy model in `mathematica/pair_displacement_check.wls`.

2.3 Squeezed limit: dilation, shear and the amplitude degeneracy

For lens modes much larger than the forest modes, $L \ll l$, expanding (4) to second order in L gives (verified in `mathematica/lensing_response_checks.wls`, check A)

$$f_{\kappa}(\mathbf{l}, \mathbf{L} - \mathbf{l}) \simeq 2\mathcal{P}(l) + 2(\hat{\mathbf{L}} \cdot \hat{\mathbf{l}})^2 l \mathcal{P}'(l) = \underbrace{[2\mathcal{P} + l\mathcal{P}']}_{\text{dilation}} + \underbrace{l\mathcal{P}' \cos 2\varphi_{lL}}_{\text{shear}}, \quad (6)$$

with $\mathcal{P}' = \partial\mathcal{P}/\partial l$ at fixed $k_{\parallel}$. The two pieces have a clean geometric meaning. For a pair separation $\boldsymbol{\theta}_{ab}$ small compared with the lens wavelength, $\boldsymbol{\alpha}_a - \boldsymbol{\alpha}_b \simeq (\boldsymbol{\theta}_{ab} \cdot \nabla) \nabla\phi = -(\kappa\mathbb{I} + \boldsymbol{\gamma})\boldsymbol{\theta}_{ab}$, so the pair correlation is evaluated at the true separation $(1 - \kappa)\boldsymbol{\theta}_{ab} - \boldsymbol{\gamma}\boldsymbol{\theta}_{ab}$: the convergence isotropically dilates the transverse correlation function while the shear makes it anisotropic. In Fourier space the exact power spectrum of a field dilated and sheared by constant $\kappa, \gamma_1, \gamma_2$ is $\mathcal{P}(|M^{-T}\mathbf{l}|)/\det M$ with $M = \mathbb{I} - \kappa\mathbb{I} - \boldsymbol{\gamma}$, whose first-order expansion equals (6) (check B in the same script).

The amplitude degeneracy is now apparent. A slowly varying multiplicative modulation of the field, $\tilde{\delta} = (1 + a)\delta_F$, produces the coupling

$$\langle \tilde{\delta}(\mathbf{l}, k_{\parallel}) \tilde{\delta}(\mathbf{l}', k'_{\parallel}) \rangle \supset 2\pi\delta_D(k_{\parallel} + k'_{\parallel}) f_a(\mathbf{l}, \mathbf{l}'; k_{\parallel}) a(\mathbf{L}), \quad f_a = \mathcal{P}(l, k_{\parallel}) + \mathcal{P}(l', k'_{\parallel}) \xrightarrow{L \ll l} 2\mathcal{P}, \quad (7)$$

so the dilation term $2\mathcal{P} + l\mathcal{P}' = (2+n)\mathcal{P}$, with $n = d \ln \mathcal{P} / d \ln l$, is proportional to the amplitude response. For a transverse spectrum that is locally a power law the convergence is therefore *indistinguishable* from a local change of the forest power spectrum amplitude by $\delta \ln P_F = (2+n)\kappa$; only the shear term $l\mathcal{P}' \cos 2\varphi$ and departures from a power law break the degeneracy. For $n = -2$ the dilation response vanishes identically, and for a spectrum that is flat in l ($n = 0$, which is what the forest looks like transversely whenever $l/\chi_* \ll k_\parallel$) there is no shear response at all and the entire lensing signal is an amplitude change. This matters because the forest power spectrum really is modulated by long-wavelength density and tidal modes, with responses measured by Chiang et al. [16], and those long modes are correlated with κ_{CMB} . That correlation is exactly the $P_F \times \kappa_{\text{CMB}}$ bispectrum detected by Doux et al. [12], Karaçaylı et al. [14], and it will enter our cubic estimator as an additive bias unless projected out (Section 2.7).

2.4 The optimal quadratic estimator for discrete pixels

Let $d_i = \tilde{\delta}(\boldsymbol{\theta}_{a(i)}, \chi_{p(i)})$, $i = 1 \dots N_{\text{pix}}$, be the vector of all forest pixels on all sightlines, in the zero-noise limit. Expand the potential in a set of real basis functions, $\phi(\boldsymbol{\theta}) = \sum_\mu p_\mu \psi_\mu(\boldsymbol{\theta})$; in the continuum limit below the ψ_μ are Fourier modes, while on a finite irregular footprint one may use Legendre or Zernike polynomials as in Metcalf et al. [3]. To first order in p the pixel covariance is, from (5),

$$C_{ij}(p) = C_{ij}^0 + \sum_\mu p_\mu C_{ij}^\mu, \quad C_{ij}^\mu = \nabla_\theta \xi(\boldsymbol{\theta}_{ab}, \Delta \chi_{pq}) \cdot [\nabla \psi_\mu(\boldsymbol{\theta}_a) - \nabla \psi_\mu(\boldsymbol{\theta}_b)]. \quad (8)$$

For a Gaussian likelihood $-2 \ln \mathcal{L} = d^T C^{-1} d + \ln \det C$ the Newton–Raphson step from $p = 0$ is the optimal (minimum-variance, unbiased to first order) quadratic estimator,

$$\hat{p}_\mu = \frac{1}{2} \sum_\nu F_{\mu\nu}^{-1} \left[d^T C_0^{-1} C^{\cdot\nu} C_0^{-1} d - \text{Tr}(C_0^{-1} C^{\cdot\nu}) \right], \quad F_{\mu\nu} = \frac{1}{2} \text{Tr}(C_0^{-1} C^{\cdot\mu} C_0^{-1} C^{\cdot\nu}). \quad (9)$$

Writing $\hat{p}_\mu = \sum_{ij} E_{ij}^\mu d_i d_j - b_\mu$ identifies the quadratic weights and the mean field,

$$E^\mu = \frac{1}{2} \sum_\nu F_{\mu\nu}^{-1} C_0^{-1} C^{\cdot\nu} C_0^{-1}, \quad b_\mu = \sum_{ij} E_{ij}^\mu C_{ij}^0. \quad (10)$$

Using $\langle d_i d_j \rangle = C_{ij}(p)$ one checks $\langle \hat{p}_\mu \rangle = p_\mu + O(p^2)$, and for Gaussian d the covariance at $p = 0$ is $\text{Cov}(\hat{p}_\mu, \hat{p}_\nu) = F_{\mu\nu}^{-1}$, which saturates the Cramér–Rao bound. The convergence coefficients follow by applying $-\nabla^2/2$ to the basis; for Fourier modes $\hat{\kappa}_L = (L^2/2)\hat{\phi}_L$. Two remarks. First, the same formulae hold with pixel noise by replacing $C^0 \rightarrow C^0 + N$, so nothing in this section is specific to the sample-variance limit except the numerical value of F . Second, because E^μ is built from ratios of C^0 and its derivative, an overall rescaling of the forest power spectrum leaves the estimator and its variance unchanged: in the zero-noise limit the reconstruction noise depends only on the *shape* of P_F and on the geometry of the pixel set.

2.5 The cubic cross-correlation estimator

Let m_k be the pixels of a CMB convergence map and $\kappa_L^{\text{CMB}} = \sum_k Y_{Lk} m_k$ its (masked, apodized) flat-sky harmonic transform, normalized so that $\langle \kappa_L \kappa_{L'}^* \rangle = \Omega \delta_{LL'} C_L$ for a full-sky-normalized spectrum. Choosing Fourier modes as the basis in (9) and converting the potential weights to convergence weights, $E^{\kappa, L} = (L^2/2)E^{\phi, L}$ and $b_L^\kappa = (L^2/2)b_L^\phi$, the natural bandpower estimator for the cross-spectrum in a bin b containing N_b modes on a periodic patch is

$$\hat{C}_b = \frac{1}{\Omega N_b} \sum_{L \in b} \text{Re}[\hat{\kappa}_L \kappa_L^{\text{CMB}*}] = \sum_{ijk} W_{ijk}^b d_i d_j m_k - \frac{1}{\Omega N_b} \sum_{L \in b} \text{Re}[b_L \kappa_L^{\text{CMB}*}], \quad W_{ijk}^b = \frac{1}{\Omega N_b} \sum_{L \in b} \text{Re}[E_{ij}^{\kappa, L} Y_{Lk}^*]. \quad (11)$$

This is the cubic form $W_{ijk}\delta_F^i\delta_F^j\kappa_{\text{CMB}}^k$ anticipated in the task description, with the weights fixed by (10). On a masked sky the harmonic coefficients are not diagonal and the bandpower expectation must be corrected with the mask mixing (MASTER) matrix; (11) is the periodic-patch form. Its properties:

- **Expectation.** To first order in ϕ , $\langle\hat{C}_b\rangle = \bar{C}_b^{\kappa_{\text{Ly}\alpha}\kappa_{\text{CMB}}}$, the bin average of the true cross-spectrum. The mean-field term is a fixed function of the pixel geometry and is uncorrelated with κ_{CMB} , so it averages to zero but should be subtracted to avoid excess variance.
- **No Gaussian ($N^{(0)}$) bias.** The disconnected part of $\langle d_i d_j m_k \rangle$ vanishes because $\langle m_k \rangle = 0$ and the Gaussian part of $d_i d_j$ is independent of m_k . This is the main advantage of the cross-correlation over the auto-spectrum of $\hat{\kappa}$, whose $N^{(0)}$ bias equals the full reconstruction noise.
- **Higher-order lensing corrections.** For Gaussian ϕ and a forest independent of ϕ , the $O(\phi^2)$ term of $\langle\tilde{\delta}\tilde{\delta}\kappa_{\text{CMB}}\rangle$ contains $\langle\phi\phi\kappa_{\text{CMB}}\rangle$, which vanishes, so there is no $N^{(1)}$ -type bias at second order in the cross-correlation. The leading Gaussian corrections to the cross-response appear at third order in the remapping expansion, schematically $C^{\phi\phi}C^{\phi\kappa_{\text{CMB}}}$, and a second-order term arises only from the lensing bispectrum or from non-Gaussian correlations between the forest and the lenses. These have not been computed here.
- **Intrinsic clustering bias.** δ_F is itself a biased tracer of the matter at $z \simeq 2-3$, which lies inside the CMB lensing kernel. The connected bispectrum $\langle\delta_F\delta_F\kappa_{\text{CMB}}\rangle_c$ therefore contributes even in the absence of lensing of the forest. Its squeezed part is the response of P_F to long modes, $\delta \ln P_F = R_\delta \delta_L + R_K K_L$ (density and tidal responses), correlated with κ_{CMB} ; in the language of Section 2.3 it enters as $\langle\hat{a}_L\kappa_L^{\text{CMB}*}\rangle \neq 0$ projected onto $\hat{\kappa}$ through the response ratio $F_{\kappa a}/F_{\kappa\kappa}$. This is the forest analogue of the gravitational-nonlinearity bias of Foreman et al. [24] and it is not small: it is the signal that Karaçaylı et al. [14] detected at 4.8σ in DESI Y1.
- **Variance.** For Gaussian fields and optimal weights the variance in the continuum limit is

$$\text{Var}(\hat{C}_b) = \frac{(C_L^{\kappa_{\text{Ly}\alpha}\kappa_{\text{Ly}\alpha}} + N_\kappa(L))(C_L^{\kappa_{\text{CMB}}\kappa_{\text{CMB}}} + N_L^{\text{CMB}}) + (C_L^{\kappa_{\text{Ly}\alpha}\kappa_{\text{CMB}}})^2}{f_{\text{sky}}(2L+1)\Delta L}, \quad (12)$$

with $N_\kappa(L)$ the reconstruction noise derived next. For a general pixel set the numerator's first factor is replaced by the appropriate combination of F^{-1} , i.e. by $\text{Cov}(\hat{\kappa}_L, \hat{\kappa}_{L'})$ from (9).

2.6 Continuum limit: Hu–Okamoto form and reconstruction noise

For infinitely dense sightlines and complete radial coverage over a depth D , the field is measured on a regular three-dimensional grid and the Fisher matrix of Fourier modes becomes diagonal. It is then simpler to derive the estimator directly. Take

$$\hat{\kappa}(\mathbf{L}) = A(L) \int \frac{dk_\parallel}{2\pi} \int \frac{d^2l}{(2\pi)^2} g(\mathbf{l}, \mathbf{L} - \mathbf{l}; k_\parallel) \tilde{\delta}(\mathbf{l}, k_\parallel) \tilde{\delta}(\mathbf{L} - \mathbf{l}, -k_\parallel). \quad (13)$$

Inserting (4), the coincident radial delta function gives $2\pi\delta_D(0) = D$, so $\langle\hat{\kappa}(\mathbf{L})\rangle = \kappa(\mathbf{L})$ requires

$$A(L)^{-1} = D \int \frac{dk_\parallel}{2\pi} \int \frac{d^2l}{(2\pi)^2} g f_\kappa. \quad (14)$$

The Gaussian variance of (13) at $\phi = 0$ has two Wick pairings, each producing another factor D and the power spectra of the two legs; minimizing it subject to the normalization gives the

standard weight $g = f_\kappa/[2\mathcal{P}(l, k_\parallel)\mathcal{P}(|\mathbf{L} - \mathbf{l}|, k_\parallel)]$ and

$$\boxed{\frac{1}{N_\kappa(L)} = D \int_{-\infty}^{\infty} \frac{dk_\parallel}{2\pi} I_{\kappa\kappa}(L, k_\parallel), \quad I_{XY}(L, k_\parallel) = \int \frac{d^2l}{(2\pi)^2} \frac{f_X(\mathbf{l}, \mathbf{L} - \mathbf{l}; k_\parallel) f_Y(\mathbf{l}, \mathbf{L} - \mathbf{l}; k_\parallel)}{2\mathcal{P}(l, k_\parallel) \mathcal{P}(|\mathbf{L} - \mathbf{l}|, k_\parallel)}} \quad (15)$$

with $\langle \hat{\kappa}(\mathbf{L}) \hat{\kappa}^*(\mathbf{L}') \rangle = (2\pi)^2 \delta_D^2(\mathbf{L} - \mathbf{L}') [C_L^{\kappa\kappa} + N_\kappa(L)]$. Equation (15) is the inverse-variance sum over independent $k_\parallel$ screens of the Hu–Okamoto noise of each screen, $N_{k_\parallel}^{-1} = I_{\kappa\kappa}$, with the mode density $D dk_\parallel/2\pi$ counting the screens [20]. It is the same object as F^{-1} of (9) per unit solid angle. Two consistency checks: (i) since $f_\kappa \propto \mathcal{P}$, the integrand is invariant under $\mathcal{P} \rightarrow c\mathcal{P}$, as it must be in the sample-variance limit; (ii) for a white spectrum, $\mathcal{P} = \text{const}$, $f_\kappa = 2\mathcal{P}$ exactly, so $I_{\kappa\kappa} = l_{\text{max}}^2/2\pi$ and $N_\kappa^{-1} = D k_{\text{max}} l_{\text{max}}^2/2\pi^2$; this equals twice the number of real degrees of freedom per steradian of the field, which is what one obtains directly by noting that a dilation by κ changes the variance of every independent white-noise pixel by a factor $(1+2\kappa)$ and each pixel then carries Fisher information $\frac{1}{2}(2)^2 = 2$. The numerical implementation (`code/recon_noise.py`) reproduces this count to the accuracy of the $|\mathbf{L} - \mathbf{l}| \leq l_{\text{max}}$ edge exclusion.

The cutoffs l_{max} and $k_{\parallel, \text{max}}$ are where the real survey enters even in this idealization: transverse modes are only measured up to the sightline Nyquist frequency, $l_{\text{max}} \simeq \pi/\theta_{\text{sep}} = \pi\sqrt{n_q}$ for n_q sightlines per steradian, radial modes up to a fraction of the pixel Nyquist frequency, and radial modes below $k_{\parallel, \text{min}} \sim 0.03 h \text{ Mpc}^{-1}$ are removed by continuum fitting. With sightlines of mean forest length D_f covering only part of the total depth, the product $D l_{\text{max}}^2$ that controls (15) is fixed by the total sightline length per unit area, so one may use $D = D_f$ together with the full n_q . Sparse sampling also aliases small-scale transverse power into the measured band; that effect is beyond the continuum treatment and is one reason the numbers below are optimistic.

2.7 Hardening against amplitude modulation

An estimator $\hat{a}(\mathbf{L})$ for the amplitude modulation is built from (13) with f_a in place of f_κ , and the two estimators respond to each other's field with response functions $R_{\kappa a}(L) = F_{\kappa a}/F_{\kappa\kappa}$ and $R_{a\kappa} = F_{\kappa a}/F_{aa}$, where $F_{XY} = D \int dk_\parallel/2\pi I_{XY}$. The bias-hardened estimator [27]

$$\hat{\kappa}^{\text{BH}} = \frac{\hat{\kappa} - R_{\kappa a} \hat{a}}{1 - R_{\kappa a} R_{a\kappa}}, \quad N_\kappa^{\text{BH}}(L) = \frac{N_\kappa(L)}{1 - \rho_{\kappa a}^2(L)}, \quad \rho_{\kappa a}^2 = \frac{F_{\kappa a}^2}{F_{\kappa\kappa} F_{aa}}, \quad (16)$$

has zero response to any $a(\boldsymbol{\theta})$ that multiplies all $k_\parallel$ modes equally. Since the true responses $R_\delta(k_\parallel, k_\perp)$ of the forest are scale dependent, a more conservative choice hardens each $k_\parallel$ screen separately, $1/N_\kappa^{\text{BH, slice}} = D \int dk_\parallel/2\pi I_{\kappa\kappa}(1 - \rho_{\kappa a}^2(L, k_\parallel))$; this removes any modulation of the form $a(\boldsymbol{\theta}, k_\parallel)$ and keeps only the information that is genuinely anisotropic or shape-changing within a screen. Both variants are evaluated below; the truth for the real forest lies in between and also requires hardening against the tidal response, which mimics shear rather than convergence and would need the $k_\perp$ -dependence of R_K to be separated from $l\mathcal{P}'$.

2.8 The signal and its detectability

In the Limber approximation, with a single forest source plane at χ_* and the CMB at χ_{CMB} ,

$$C_L^{XY} = \int_0^{\chi_{\text{min}}} d\chi \frac{W_X(\chi) W_Y(\chi)}{\chi^2} P_{\text{m}}\left(k = \frac{L+1/2}{\chi}, z(\chi)\right), \quad W_s(\chi) = \frac{3}{2} \Omega_{\text{m}} \left(\frac{H_0}{c}\right)^2 (1+z) \chi \frac{\chi_s - \chi}{\chi_s}, \quad (17)$$

and the total detection significance of the cross-spectrum from (12) is

$$\left(\frac{S}{N}\right)^2 = \sum_L f_{\text{sky}} (2L+1) \frac{(C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\text{CMB}}})^2}{(C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\text{Ly}\alpha}} + N_\kappa(L)) (C_L^{\kappa_{\text{CMB}} \kappa_{\text{CMB}}} + N_L^{\text{CMB}}) + (C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\text{CMB}}})^2}. \quad (18)$$

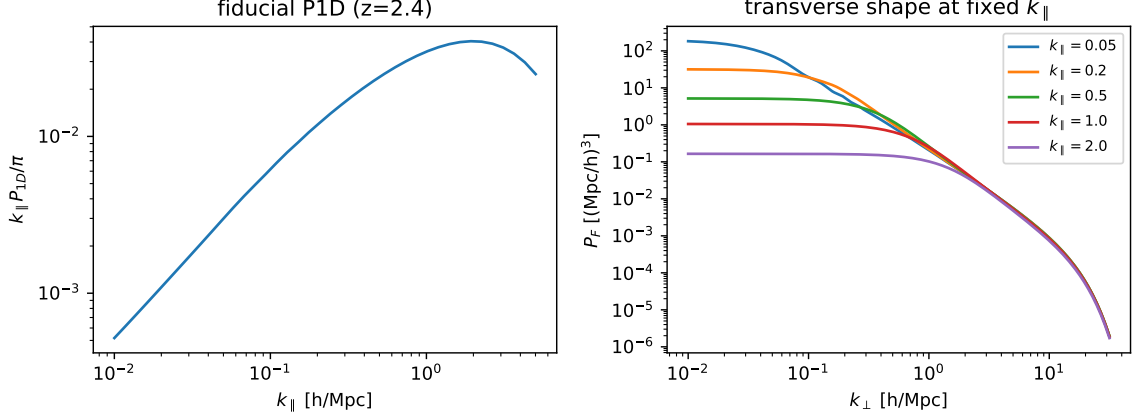


Figure 1: Fiducial forest power spectrum at $z = 2.4$. Left: the implied one-dimensional power. Right: the transverse shape at fixed $k_{\parallel}$; for $k_{\perp} \ll k_{\parallel}$ the spectrum is flat in l , which is the regime in which lensing is degenerate with an amplitude change (Section 2.3).

Because the forest source plane sits well inside the CMB lensing kernel the two convergences are strongly correlated, with $r_L = C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\text{CMB}}} / (C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\text{Ly}\alpha}} C_L^{\kappa_{\text{CMB}} \kappa_{\text{CMB}}})^{1/2} \simeq 0.9$ at $L = 10$ falling to 0.67 at $L \gtrsim 300$.

3 Toy numbers

Fiducial model. The forest power spectrum is the Kaiser form with the non-linear correction of Arinyo-i-Prats et al. [31], $P_F = b_F^2 (1 + \beta_F \mu^2)^2 P_{\text{lin}}(k) D_{\text{NL}}(k, \mu)$, with approximate $z = 2.4$ parameters $b_F = -0.13$, $\beta_F = 1.6$, $q_1 = 0.6$, $q_2 = 0$, $k_v = 1 \text{ h Mpc}^{-1}$, $a_v = 0.55$, $b_v = 1.6$, $k_p = 16 \text{ h Mpc}^{-1}$ (`code/forest_power.py`; only the shape matters here). The linear and non-linear matter power spectra come from CAMB for a Planck-2018 cosmology. The radial depth is $D = 350 \text{ h}^{-1} \text{ Mpc}$ (one forest), the radial band is $0.03 \leq k_{\parallel} \leq k_{\parallel, \text{max}}$ with $k_{\parallel, \text{max}} \in \{1, 2, 5\} \text{ h Mpc}^{-1}$, and the transverse cutoff is the sightline Nyquist multipole for $n_q \in \{20, 50, 100, 400\} \text{ deg}^{-2}$ ($l_{\text{max}} = 1273$ for the DESI-DR1-like value $n_q = 50$; DR1 has about 45 forests per deg^2 over 9500 deg^2). Figure 1 shows the fiducial spectrum.

CMB lensing noise. Rather than fabricate noise curves, we use white noise levels calibrated to reproduce the published detection significances of the auto-spectra: $N_L^{\text{CMB}} = 1.68 \times 10^{-7}$ reproduces 43σ over $40 \leq L \leq 763$ on $f_{\text{sky}} = 0.23$ for ACT DR6 [6], and $N_L^{\text{CMB}} = 3.03 \times 10^{-7}$ reproduces the $\simeq 42\sigma$ -equivalent amplitude error over $8 \leq L \leq 400$ on $f_{\text{sky}} = 0.67$ for Planck PR4 [8]. Real noise curves rise steeply at high L and should replace these before any forecast is quoted externally. Overlap fractions of $f_{\text{sky}} = 0.15$ (ACT) and 0.22 (Planck, the full DESI DR1 forest area) are assumed.

Results. Figure 2 compares $N_{\kappa}(L)$ with the signal spectra and Table 1 lists values at $L = 100$, where $C_{100}^{\kappa_{\text{Ly}\alpha} \kappa_{\text{Ly}\alpha}} = 2.78 \times 10^{-8}$, $C_{100}^{\kappa_{\text{Ly}\alpha} \kappa_{\text{CMB}}} = 4.57 \times 10^{-8}$ and $C_{100}^{\kappa_{\text{CMB}} \kappa_{\text{CMB}}} = 1.38 \times 10^{-7}$.

Three features of these numbers are robust even though their absolute values are not:

1. For a DESI-like density the naive sample-variance noise at $L \sim 100$, $N_{\kappa} = 2.45 \times 10^{-8}$, is comparable to the signal $C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\text{Ly}\alpha}}$, and it falls steeply with $k_{\parallel, \text{max}}$ (as $1/k_{\parallel, \text{max}}$ when the transverse spectrum is flat) and with n_q (as $1/l_{\text{max}}^2 \propto 1/n_q$).
2. The amplitude-hardened noise is ~ 8 times larger for $n_q = 50$ ($N_{\kappa}^{\text{BH}} = 1.98 \times 10^{-7}$), ~ 17 times larger for $n_q = 20$, and it is almost independent of $k_{\parallel, \text{max}}$. This is the quantitative

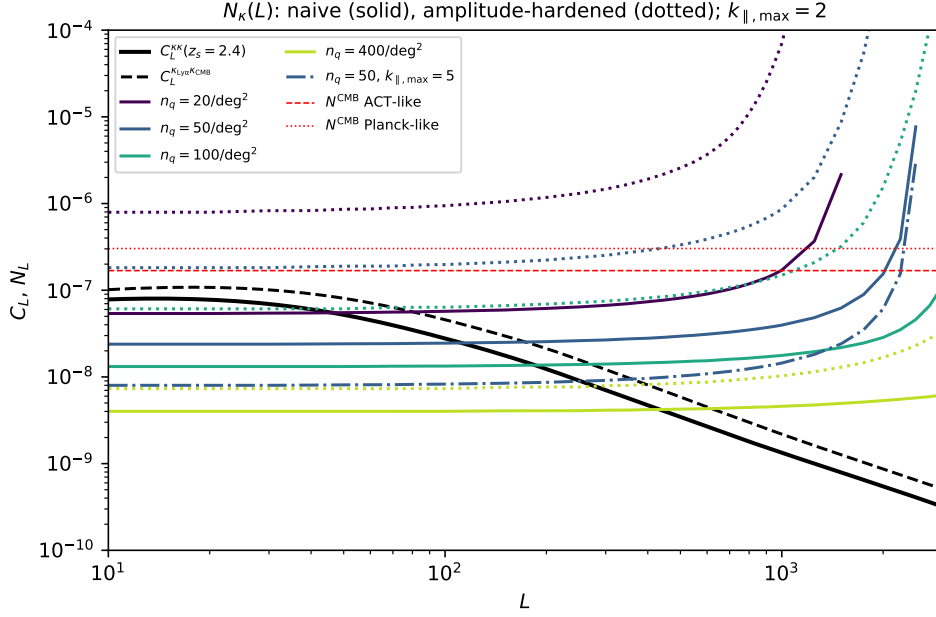


Figure 2: Sample-variance-limited reconstruction noise $N_\kappa(L)$ (solid, Eq. 15) for sightline densities n_q and $k_{\parallel,\max} = 2 h \text{ Mpc}^{-1}$, its per-screen amplitude-hardened version (dotted, Eq. 16), and the case $k_{\parallel,\max} = 5$ for $n_q = 50$ (dash-dotted). Black curves are the forest convergence auto-spectrum and its cross-spectrum with κ_{CMB} ; red lines are the white-noise equivalents of the ACT DR6 and Planck PR4 reconstruction noise.

Table 1: Reconstruction noise at $L = 100$ and cumulative cross-correlation signal-to-noise (to $L = 3000$) in the sample-variance limit. N^{BH} is the per-screen amplitude-hardened noise. S/N columns are for $k_{\parallel,\max} = 2$; the last column uses N^{BH} with ACT-like CMB noise.

$n_q [\text{deg}^{-2}]$	$l_{\max}$	$k_{\parallel,\max}$	$N_\kappa(100)$	$N_\kappa^{\text{BH}}(100)$	S/N ACT	S/N Planck	S/N ACT, BH
20	805	1.0	1.36×10^{-7}	9.49×10^{-7}			
		2.0	5.71×10^{-8}	9.44×10^{-7}	25.7	25.4	7.1
		5.0	2.04×10^{-8}	9.43×10^{-7}			
50	1273	1.0	6.21×10^{-8}	2.01×10^{-7}			
		2.0	2.45×10^{-8}	1.98×10^{-7}	36.5	35.7	15.2
		5.0	8.27×10^{-9}	1.97×10^{-7}			
100	1800	1.0	3.40×10^{-8}	6.63×10^{-8}			
		2.0	1.33×10^{-8}	6.37×10^{-8}	46.1	44.6	25.0
		5.0	4.26×10^{-9}	6.32×10^{-8}			
400	3600	1.0	8.15×10^{-9}	8.84×10^{-9}			
		2.0	4.01×10^{-9}	7.33×10^{-9}	67.4	64.2	54.4
		5.0	1.21×10^{-9}	7.05×10^{-9}			

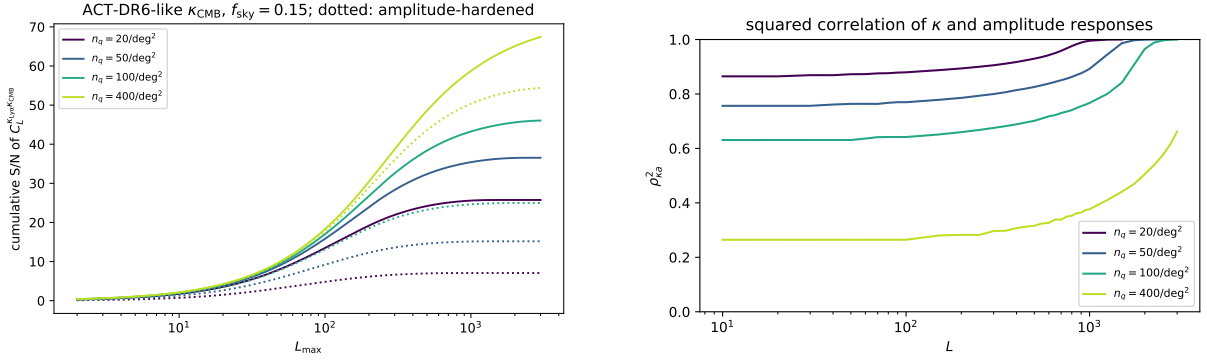


Figure 3: Left: cumulative signal-to-noise of $C_L^{K_{Ly\alpha} K_{CMB}}$ for ACT-like CMB noise and $f_{\text{sky}} = 0.15$, with (dotted) and without (solid) amplitude hardening. Right: squared correlation coefficient between the convergence and amplitude responses, $\rho_{\kappa a}^2$ of Eq. (16), after summing over $k_{\parallel}$ screens.

form of Section 2.3: radial modes with $k_{\parallel} \gg l_{\max}/\chi_*$ see a transversely flat spectrum, and for them lensing is purely an amplitude change. The information that survives hardening comes from the few screens with $k_{\parallel} \lesssim l_{\max}/\chi_* \simeq 0.3 h \text{ Mpc}^{-1}$, so it is set by the sightline density alone. Increasing the density from 50 to 400 deg^{-2} reduces N_{κ}^{BH} by a factor ~ 30 , faster than the n_q^{-1} scaling of the naive noise, because it also converts amplitude-degenerate modes into shear-sensitive ones.

3. With ACT-like CMB noise the sample-variance-limited cross-correlation would be detected at $\simeq 37\sigma$ for $n_q = 50$ (Planck: 36σ), dominated by $L \lesssim 500$. These numbers are upper limits of no practical relevance by themselves: DESI pixel noise ($\sigma_{\delta} \simeq 0.8$ per $0.8 \text{ \AA}$ pixel against a forest rms of 0.34) removes most radial modes above $k_{\parallel} \sim 0.5 h \text{ Mpc}^{-1}$, sparse sampling aliases transverse power, and the estimator's own $N^{(1)}$ and clustering biases are not included. Shaw et al. [5] find $S/N \simeq 3\text{--}4$ for the reconstructed potential against foreground galaxies with realistic noise. The route from the numbers here to a realistic forecast is to add the pixel noise to $\mathcal{P}$ in (15), treat the sightline sampling explicitly through (9), and compute the bias terms.

4 Three tracers: separating lensing from the forest response

Section 3 showed that hardening the estimator against an amplitude modulation costs most of the sample-variance information. This section asks a different question: with three fields on the same footprint, the quasars q at the forest redshift, the forest δ_F and κ_{CMB} , which power spectra and bispectra separate the lensing of the forest from its response to long-wavelength density and tidal modes without assuming that response is known? The answer is formulated in the squeezed limit and at the level of spectra only.

4.1 Squeezed-limit model

Let $\hat{P}_F(\mathbf{k}|\boldsymbol{\theta})$ be the forest power spectrum measured locally in a cell around $\boldsymbol{\theta}$ over the redshift slab of interest (in practice the pixel-pair estimator of Section 2.4 restricted to a cell, or the position-dependent power spectrum). Its fractional fluctuation, $p(\mathbf{k};\boldsymbol{\theta}) = \delta \ln \hat{P}_F(\mathbf{k}|\boldsymbol{\theta})$, is a two-dimensional field for every short-mode bin $\mathbf{k} = (k_{\parallel}, k_{\perp}, \varphi)$, with φ the angle of $\mathbf{k}_{\perp}$ to the long-mode direction $\hat{\mathbf{L}}$. To first order in the long modes,

$$p(\mathbf{k}; \mathbf{L}) = T(\mathbf{k}) \kappa_{\text{Ly}\alpha}(\mathbf{L}) + \mathcal{R}(\mathbf{k}) \delta_s(\mathbf{L}) + \dots, \quad (19)$$

where δ_s is the matter density projected over the slab and:

- $T(\mathbf{k}) = (2 + n_\perp) + n_\perp \cos 2\varphi$, with $n_\perp = \partial \ln P_F / \partial \ln k_\perp$ at fixed $k_\parallel$, is the lensing template of Eq. (6). It is *known*, because it is built from the measured P_F itself.
- $\mathcal{R}(\mathbf{k}) = R_\delta(k_\parallel, k_\perp) + R_K(k_\parallel, k_\perp) \cos 2\varphi$ is the response of the forest to the long mode. For a long mode with $k_\parallel \simeq 0$, which is the only kind that projects onto a two-dimensional field such as κ_{CMB} , the tidal tensor is fixed by the density, $K_{ij} = (\hat{L}_i \hat{L}_j - \delta_{ij}/3) \delta_s$, so the tidal response is not an independent field: it is a $\cos 2\varphi$ harmonic multiplying the same $\delta_s(\mathbf{L})$ (`mathematica/three_tracer_checks.wls`, check 5). $\mathcal{R}$ is treated as *unknown*: one number per short-mode bin and harmonic.

Both terms have the same angular structure (an $m = 0$ and an $m = 2$ harmonic), so they cannot be separated by the angular dependence alone. They can be separated because they multiply different long-mode fields: $\kappa_{\text{Ly}\alpha}$ is sourced by matter at $z < z_s$, δ_s by matter at $z \simeq z_s$, and the quasars trace the latter and not the former.

The quasar overdensity projected over the same slab is

$$q(\mathbf{L}) = b_q \delta_s(\mathbf{L}) + m \kappa_q(\mathbf{L}) + \epsilon(\mathbf{L}), \quad m \equiv 5s - 2, \quad (20)$$

with $b_q \simeq 3.5$ the quasar bias, $m \kappa_q$ the magnification-bias term ($\kappa_q \simeq \kappa_{\text{Ly}\alpha}$ since the quasars sit at the forest redshift) and ϵ shot noise plus any stochasticity.

4.2 The measurements and the deprojection identity

Cross-correlating p with the two long-mode fields gives two families of bispectra (they are $\langle \delta_F \delta_F X \rangle$ in squeezed configurations, with the short modes binned in $\mathbf{k}$):

$$B_{FF\kappa}(\mathbf{k}; L) \equiv \langle p(\mathbf{k}) \kappa_{\text{CMB}} \rangle_L = \mathcal{R}(\mathbf{k}) C_L^{\delta_s \kappa_{\text{CMB}}} + T(\mathbf{k}) C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\text{CMB}}}, \quad (21)$$

$$B_{FFq}(\mathbf{k}; L) \equiv \langle p(\mathbf{k}) q \rangle_L = \mathcal{R}(\mathbf{k}) C_L^{\delta_s q} + T(\mathbf{k}) C_L^{\kappa_{\text{Ly}\alpha} q}. \quad (22)$$

B_{FFq} is the quasar–forest-power cross-correlation proposed by Chiang et al. [16]; $B_{FF\kappa}$ integrated over $\mathbf{k}_\perp$ is the $P_{1D}-\kappa_{\text{CMB}}$ bispectrum of Doux et al. [12], Karaçaylı et al. [14], for which T vanishes identically because a single skewer is invariant under a transverse remapping (the dilation response integrates to a boundary term, $\int_0^{l_{\text{max}}} (2\mathcal{P} + l\mathcal{P}') l dl = l_{\text{max}}^2 \mathcal{P}(l_{\text{max}})$, check 4). Form the linear combination

$$\boxed{\mathcal{D}(\mathbf{k}; L) \equiv B_{FF\kappa}(\mathbf{k}; L) - \beta_L B_{FFq}(\mathbf{k}; L) = T(\mathbf{k}) \left[C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\text{CMB}}} - \beta_L C_L^{\kappa_{\text{Ly}\alpha} q} \right], \quad \beta_L \equiv \frac{C_L^{\delta_s \kappa_{\text{CMB}}}}{C_L^{\delta_s q}}} \quad (23)$$

in which the unknown response cancels identically for every $\mathbf{k}$ (check 1). Equivalently, one cross-correlates the full quadratic estimator $\hat{\kappa}$ of Section 2.6 with the deprojected CMB map $\kappa_\perp \equiv \kappa_{\text{CMB}} - \beta q$.

Kernel matching. Equation (23) holds under the model (19), in which the response multiplies a single projected density. In the measurement the response of a short-mode bin is a radial integral $p_\alpha^{\text{int}} = \int d\chi A_\alpha(\chi) \delta(\chi)$ with bin-dependent weights A_α set by forest evolution, coverage, noise and continuum fitting. The residual after deprojection is then

$$\int d\chi A_\alpha(\chi) \left[W_{\text{CMB}}(\chi) - \beta_L b_q(\chi) W_q(\chi) \right] \frac{P((L + \frac{1}{2})/\chi, \chi)}{\chi^2}, \quad (24)$$

which vanishes for arbitrary A_α only if the bracket vanishes pointwise in χ . A single slab-averaged β with a top-hat quasar kernel does not achieve this: W_{CMB} rises from 3.61×10^{-4} to $4.86 \times 10^{-4} h \text{ Mpc}^{-1}$ across the $2.1 < z < 3.0$ slab (`mathematica/pair_template_checks.wls`,

check 8). The fix is to build the quasar template with a radial weight that matches the CMB kernel,

$$\hat{\kappa}_s(\boldsymbol{\theta}) = \int_{\text{slab}} d\chi W_{\text{CMB}}(\chi) \frac{\delta_q(\boldsymbol{\theta}, \chi)}{b_q(\chi)} = \sum_{i \in \boldsymbol{\theta}} \frac{W_{\text{CMB}}(\chi_i)}{b_q(\chi_i) \bar{n}_{3D}(\chi_i) \chi_i^2} - \text{mean}, \quad (25)$$

an estimate of the contribution of the slab itself to κ_{CMB} . Then $\beta \equiv 1$, the bracket in (24) vanishes identically for any radial response weight, and the only inputs are $W_{\text{CMB}}(\chi)$ (geometry) and $b_q(z)$, which DESI measures to a few per cent from the quasar auto-correlation. A fractional error ϵ in b_q leaves a fraction ϵ of the contamination: the residual is first order in the coefficient error, not second order. The shot noise of the weighted template is $D^2 \langle W_{\text{CMB}}^2 \rangle / (b_q^2 \bar{n}_q) = 8.99 \times 10^{-8}$, marginally above the $\beta^2 / \bar{n}_q = 8.58 \times 10^{-8}$ of the single-coefficient version.

What still has to hold.

1. **Locality in χ .** The forest response must be to the density at the same χ ; a long mode with $k_{\parallel} \neq 0$ within the slab window ($k_{\parallel} \lesssim 1/D$) has a tidal tensor that differs from the transverse one at $O(k_{\parallel}^2/k^2)$ and a quasar redshift-space distortion at $O(\mu^2)$; both are suppressed for the $k_{\parallel} \lesssim 0.01 h \text{ Mpc}^{-1}$ modes that project onto a two-dimensional field, but the residual radial transfer mismatch between density, tides and quasar RSD must be tested on mocks rather than assumed.
2. **Deterministic quasar bias.** With $b_q(z)$ imposed rather than fitted, Poisson noise and any stochasticity ϵ uncorrelated with the forest response add variance, not bias; a component of ϵ correlated with the forest response, or a scale-dependent bias not captured by $b_q(z)$, biases the residual by the corresponding cross-spectrum. (The $(1 - r_{q\delta}^2)$ formula of check 2 applies to the fitted-coefficient version only.)
3. **Magnification.** With $q = b_q \delta + m \kappa_q$, the template (25) contains $(m/b_q) \int W_{\text{CMB}} \kappa_q$, a lensing term. In the single-coefficient version the exact cancellation coefficient is $\beta = C^{\delta_s \kappa_{\text{CMB}}} / (b_q C^{\delta_s \delta_s} + m C^{\delta_s \kappa_q})$ and ignoring m altogether biases β by $\Delta\beta/\beta = 9$ per cent at $L = 100$ for $m = 0.5$ (check 3b), which leaks first order into the residual. $m = 5s - 2$ must therefore be fitted or measured, most cleanly from C_L^{qq} with a low-redshift galaxy sample, which has no intrinsic overlap. Matter inside the slab also lenses the more distant forest pixels, so the template subtracts real signal: the total subtraction $C^{\kappa_{\text{Ly}\alpha} \hat{\kappa}_s} / C^{\kappa_{\text{Ly}\alpha} \kappa_{\text{CMB}}}$ is 1.1 per cent at $L = 100$ in the fiducial model, and it is kept in the signal template rather than neglected.
4. **Shared quasars.** The template quasars in a slab also provide forest pixels in the lower part of the same slab, so the local sightline density, the template shot noise, the coverage, the continuum errors and the flux-dependent weights are all correlated with q . Taking pixels only from quasars behind the whole slab would remove standard-region coverage below $z \simeq 2.4$ and is not an option. Exact mean-field subtraction handles the pure geometry; it does not guarantee $\mathbb{E}[d_i d_j | \text{selection}] = C_{ij}^0$. The first measurement therefore uses tomographic sub-slabs with disjoint template and sightline selections, monitors the sightline-density $\times \kappa_{\text{CMB}}$ cross-spectrum, and includes the shared sampling in the joint mocks; disjoint selections remove the shared-object terms but not shared-LSS or magnification correlations, which the mocks must carry.
5. **Magnification leakage.** With $m(z)$ the template contains $M(\boldsymbol{\theta}) = \int W_{\text{CMB}} (m/b_q) \kappa_q$; besides removing a little lensing signal (kept in the signal template), M correlates with the forest response through the intrinsic overlap of κ_q 's kernel with the slab, a contaminant of order the 1 per cent overlap that is modelled with the actual quasar $n(z)$ and $m(z)$ rather than neglected.

6. **Beyond Limber.** Even an exact subtraction of the slab density leaves the correlation between densities inside and outside the slab, $\int_{\text{slab}} d\chi A_\alpha(\chi) \int_{\text{outside}} d\chi' W_{\text{CMB}}(\chi') C_L^{\delta(\chi)\delta(\chi')}$, which is confined to a correlation length of the slab edges and is removed by extending the template slab beyond the forest slab by a margin of $\sim 150 h^{-1}\text{Mpc}$ on each side; and quasar redshift-space distortions add $f\mu^2/b_q$ to the template, which at $L = 40$ ($k_\perp \simeq 0.01 h \text{Mpc}^{-1}$, comparable to the radial window $\sim 1/D$) is a ~ 10 per cent effect and at $L \geq 100$ a few per cent. Both are quantified with non-Limber calculations and in the joint mocks, and set the lower multipole of the clean range.

The complete set of measurements is therefore: (i) $B_{FF\kappa}(\mathbf{k}; L)$, (ii) $B_{FF\hat{\kappa}_s}(\mathbf{k}; L)$ with the matched template, (iii) C_L^{qq} in redshift shells for $b_q(z)$, (iv) C_L^{qq} with a low- z galaxy sample for magnification, and (v) the forest correlation function that defines the lensing template. The lensing detection is the amplitude of $\mathcal{D}$ against the template; the consistency test is that $\mathcal{D}/T$ is independent of $\mathbf{k}$; and $B_{FF\hat{\kappa}_s}$ measures the response $\mathcal{R}$ itself, up to the small magnification-lensing leak. Section 5 implements the same logic without cells or harmonics.

4.3 Realistic noise

The sparse, noisy sampling of the forest enters (15) by replacing $\mathcal{P} \rightarrow \mathcal{P} + \mathcal{N}$ in the denominators, with the response f_κ unchanged (self-pairs of a sightline carry no lensing response, Eq. 8 with $a = b$, and only add variance). For Poisson-distributed sightlines with effective density $\bar{n}_{\text{eff}}$ per steradian at the redshift in question, the noise is white in l and given by the one-dimensional power of a skewer,

$$\mathcal{N}(k_\parallel) = \frac{P_N + P_{1D}(k_\parallel)}{\bar{n}_{\text{eff}}}, \quad P_N = \sigma_\delta^2 \Delta\chi_{\text{pix}}, \quad (26)$$

the form used for three-dimensional forest power spectra by McQuinn & White. The P_{1D} term is the aliasing of unresolved transverse power; it is present even for perfect spectra and dominates at DESI densities. For DESI DR1, $\sigma_\delta \simeq 0.8$ per $0.8 \text{ \AA}$ pixel ($\Delta\chi_{\text{pix}} \simeq 0.55 h^{-1}\text{Mpc}$ at $z = 2.4$) gives $P_N \simeq 0.33 h^{-1}\text{Mpc}$, to be compared with $P_{1D}(k_\parallel = 0.1) \simeq 0.19 h^{-1}\text{Mpc}$; the $P_N \simeq 0.17$ case corresponds to the cleaned sample of Shaw et al. [5]. About 45 forests per deg^2 each covering $\sim 350 h^{-1}\text{Mpc}$ of a $712 h^{-1}\text{Mpc}$ deep slab give an unweighted available-sightline density $\bar{n}_{\text{los}} \simeq 22 \text{ deg}^{-2}$ at any one redshift (this replaces the 25 used in an earlier draft; the distinction from a noise-weighted effective density matters because the harmonic-mean weighting below already accounts for the noise); 50 and 100 deg^{-2} stand for a complete DESI or a denser survey. Two limitations of (26) should be kept in mind: replacing the full sampling covariance by independent Gaussian modes with this diagonal power ignores the mode couplings induced by sparse sampling, and Poisson sampling has no sharp Nyquist frequency, so the $l_{\text{max}} = \pi\sqrt{\bar{n}}$ cutoff is an additional restriction rather than part of the noise (relaxing it to $2\pi\sqrt{\bar{n}}$ lowers $N_\kappa(100)$ by 18 per cent). Both are to be validated against the discrete estimator on mocks.

Inverse-variance weighting of sightlines. Equation (26) with a single P_N describes a homogeneous population with equal weights. Real quasars span more than two magnitudes, so P_N varies by well over an order of magnitude from sightline to sightline. The discrete estimator (9) weights every pixel by $C^{-1} = (C^0 + N)^{-1}$ automatically; in the continuum formula the same C^{-1} weighting appears because independent sightlines contribute additively to the Fisher information per mode, so that

$$\frac{1}{\mathcal{N}(k_\parallel)} = \bar{n}_{\text{eff}} \left\langle \frac{1}{P_{N,a} + P_{1D}(k_\parallel)} \right\rangle_a, \quad (27)$$

a harmonic rather than arithmetic mean over the noise distribution. The aliasing term stays inside the average because the unresolved small-scale structure of each skewer is independent noise for that skewer. We model $P_{N,a}$ as log-normal with median $0.33 h^{-1}\text{Mpc}$ and $\ln$ -scatter $\sigma_{\ln}$; since $P_N \propto \sigma_\delta^2 \propto f^{-2}$ for background-limited spectra, a factor-of-three scatter in quasar

Table 2: Reconstruction noise and cross-correlation signal-to-noise with realistic forest noise, for the slab $2.1 < z < 3.0$, $R_\delta = 2$, $b_q = 3.5$, $m = 0.5$, ACT-like CMB noise with $f_{\text{sky}} = 0.15$ and Planck-like with 0.22. “Naive” is $\langle \hat{\kappa} \kappa_{\text{CMB}} \rangle$, whose expectation is biased by the amount in the fifth column; “BH” is the per-screen hardened estimator of Section 2.7; “matched” is the deprojection with the kernel-matched quasar template (25); “response” is the significance of the response term itself. $\sigma_{\text{ln}} > 0$ rows use the inverse-variance weighting of Eq. (27).

$\bar{n}_{\text{los}}$	P_N	σ_{ln}	$N_\kappa(100)$	bias/signal at $L = 100$	all L , ACT			cumulative S/N $40 \leq L \leq 300$		
[deg $^{-2}$]	[h^{-1} Mpc]				naive	BH	matched	matched ACT	matched Pl.	response
22	0.00	0	2.43×10^{-6}	1.58	4.2	2.6	3.8	3.1	3.2	5.8
22	0.17	0	6.73×10^{-6}	1.58	2.6	1.6	2.3	1.9	2.0	3.6
22	0.33	0	1.21×10^{-5}	1.59	1.9	1.2	1.7	1.4	1.5	2.7
50	0.00	0	5.94×10^{-7}	1.46	8.1	5.8	7.2	5.7	5.9	10.0
50	0.17	0	1.64×10^{-6}	1.44	5.2	3.8	4.6	3.7	3.9	6.4
50	0.33	0	2.89×10^{-6}	1.45	4.0	2.9	3.6	2.9	3.0	5.0
100	0.00	0	1.90×10^{-7}	1.37	12.8	10.8	11.2	8.4	8.7	14.1
100	0.17	0	5.32×10^{-7}	1.31	8.8	7.1	7.8	6.1	6.3	9.6
100	0.33	0	9.28×10^{-7}	1.32	6.9	5.5	6.1	4.8	5.0	7.6
22	0.33	1	1.09×10^{-5}	1.59	2.0	1.3	1.8	1.5	1.6	2.9
22	0.33	2	9.52×10^{-6}	1.60	2.2	1.4	1.9	1.6	1.7	3.1
50	0.33	1	2.59×10^{-6}	1.45	4.2	3.0	3.7	3.0	3.1	5.2
50	0.33	2	2.25×10^{-6}	1.46	4.5	3.2	4.0	3.2	3.4	5.6
DR1 emp.	$n(z)$, $P_{N,a}$	data	3.07×10^{-5}	1.68	1.2	0.7	1.1	0.9	0.9	1.8

flux gives $\sigma_{\text{ln}} \simeq 2$, and $\sigma_{\text{ln}} = 1$ is a conservative case. Here $\bar{n}_{\text{eff}}$ is the unweighted density $\bar{n}_{\text{los}}$. Relative to the equal-weight median-noise case, (27) lowers $\mathcal{N}$ by a factor 1.1 ($\sigma_{\text{ln}} = 1$) or 1.1 ($\sigma_{\text{ln}} = 2$) at $k_{\parallel} = 0.1 h \text{ Mpc}^{-1}$, and by 1.1 or 1.4 at $k_{\parallel} = 1 h \text{ Mpc}^{-1}$ where pixel noise dominates over aliasing. The actual DR1 distribution of per-pixel noise should replace this model; it is available from the public DR1 delta files. The quasar template has $\bar{n}_q = 25 \text{ deg}^{-2}$ in the slab (shot noise $1/\bar{n}_q = 1.2 \times 10^{-5} \text{ sr}$), the CMB noise is the white-noise equivalent of Section 3 recalibrated after extending the Limber integral for $C^{\kappa_{\text{CMB}} \kappa_{\text{CMB}}}$ to recombination (the earlier $z = 6$ cap dropped 7–32 per cent of it), and the deprojected map has power $C^{\kappa_{\perp} \kappa_{\perp}} = C^{\kappa_{\text{CMB}} \kappa_{\text{CMB}}} + N^{\text{CMB}} - 2\beta C^{q \kappa_{\text{CMB}}} + \beta^2 (C^{qq} + 1/\bar{n}_q)$; for the matched template the same with $\beta = 1$, $q \rightarrow \hat{\kappa}_s$. Deprojection removes only $r_{q\kappa}^2 = 0.033$ of the CMB lensing power at $L = 100$, and the added shot-noise term is comparable to $C_{100}^{\kappa_{\text{CMB}} \kappa_{\text{CMB}}} = 1.59 \times 10^{-7}$. The variance of the forest side includes the response sample variance, $C^{\hat{\kappa} \hat{\kappa}} = C^{\kappa_{\text{Ly}\alpha} \kappa_{\text{Ly}\alpha}} + 2a C^{\kappa_{\text{Ly}\alpha} \delta_s} + a^2 C^{\delta_s \delta_s} + N_\kappa$ with $a = (R_\delta/2) R_{\kappa a}$; the factor 1/2 converts the power response $R_\delta = d \ln P_F / d\delta$ to the field modulation a to which the estimator responds.

4.4 Results

Table 2 and Figures 4–5 give the outcome. The S/N is quoted both over all L and over $40 \leq L \leq 300$, the range a cell- or pair-based measurement with a released CMB map will actually cover; the latter is the headline. Three points:

1. **The contamination is comparable to the signal.** With $R_\delta = 2$ the response term in $\langle \hat{\kappa} \kappa_{\text{CMB}} \rangle$ is 0.82 times the lensing signal at $L = 40$ and 1.63 times at $L = 100$ (an earlier draft quoted twice these values, having applied the power response R_δ where the field response $R_\delta/2$ belongs). The ratio grows with L because the slab density spectrum is shallower than the lensing spectrum, and the estimator transfers it almost fully, $R_{\kappa a} \simeq 0.8$ –1. The response term alone would be measured at 2.7σ over $40 \leq L \leq 300$ with DR1 noise: the deprojection is not optional, and the response measurement is a positive control.
2. **Sampling noise, not pixel noise, sets the scale.** Aliasing alone raises $N_\kappa(100)$ from

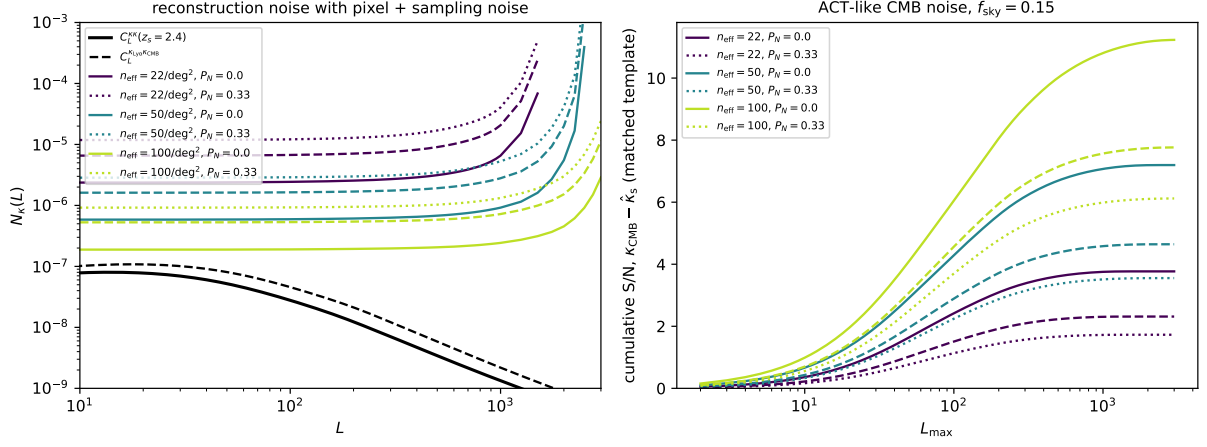


Figure 4: Left: reconstruction noise with the sampling and pixel noise of Eq. (26) (solid: perfect spectra, dotted: DESI DR1 pixel noise) for effective sightline densities of 25, 50 and 100 deg^{-2} . Right: cumulative signal-to-noise of the cross-correlation with the matched-template deprojected map, $\langle \hat{\kappa} (\kappa_{\text{CMB}} - \hat{\kappa}_s) \rangle$.

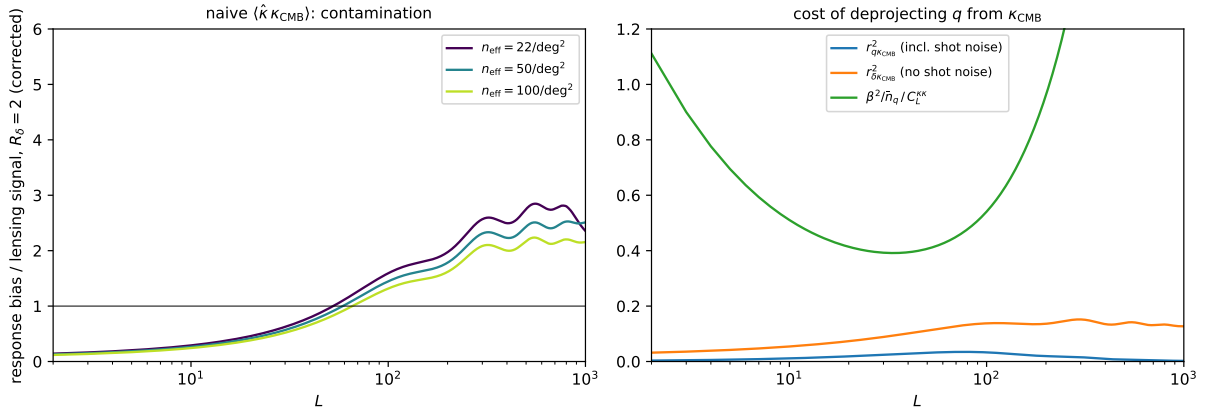


Figure 5: Left: ratio of the response contamination to the lensing signal in the naive $\langle \hat{\kappa} \kappa_{\text{CMB}} \rangle$ for a fiducial response $R_\delta = 2$, with DR1 noise. Right: the cost of deprojecting q from κ_{CMB} : the fraction of CMB lensing power removed ($r_{q\kappa}^2$) and the added quasar shot-noise term relative to $C_L^{\kappa_{\text{CMB}} \kappa_{\text{CMB}}}$.

2.5×10^{-8} (dense sampling, Section 3) to 2.43×10^{-6} at 22 deg^{-2} ; DR1 pixel noise raises it further to 1.21×10^{-5} . Only modes with $k \lesssim 0.15 h \text{ Mpc}^{-1}$ are signal dominated.

3. **What remains.** For DR1-like parameters the matched-template cross-correlation reaches $S/N = 1.4$ over $40 \leq L \leq 300$ (1.7 over all L) with ACT-like and 1.5 with Planck-like CMB noise, against 1.2 for the hardened estimator. Cleaner spectra ($P_N = 0.17$) give 1.9; inverse-variance weighting of a heterogeneous population with $\sigma_{\text{ln}} = 2$ gives 1.6; perfect spectra 3.1; a complete DESI with $\bar{n}_{\text{los}} = 50$ and clean spectra 3.7. Weighting recovers part of the gap to the perfect-spectra case but cannot cross it: the aliasing term $P_{1D}/\bar{n}$ is a floor, and it is the dominant noise at DESI densities. **Empirical DR1 inputs.** With the public DR1 deltas in hand (`code/data_checks/delta_summary.py`: 428 403 forests, $\sim 11\,000 \text{ deg}^2$ of nside-64 pixels containing at least one forest), the forest coverage falls from 22 deg^{-2} at $z = 2.1$ to 15 at $z = 2.4$, 8 at $z = 2.75$ and 4.6 at $z = 3.0$, and the per-forest noise power $P_{N,a} = \Delta\chi_{\text{pix}}/\text{MEANSNR}_a^2$ has median $0.54 h^{-1} \text{ Mpc}$ with a ln-scatter of 2.2, broader and higher than the 0.33 assumed above. Replacing the flat $\bar{n}_{\text{los}}$ and the log-normal by the measured coverage $n(z)$ and the measured $P_{N,a}$ (inverse-variance weighted, slab-averaged) gives the “DR1 empirical” row of Table 2: $S/N = 0.9$ (ACT) and 0.9 (Planck) over $40 \leq L \leq 300$, with the response term at 1.8σ . The footprint area used for the densities is an upper estimate, so these numbers are if anything slightly low. These numbers are consistent with the $S/N \simeq 3\text{--}4$ that Shaw et al. [5] find for the full DESI footprint against foreground galaxies, once scaled to their area.

The conclusion for DESI DR1 is a marginal measurement: below 2σ with the idealised flat coverage, and $\sim 1\sigma$ once the measured coverage $n(z)$ and noise distribution are used, in this Gaussian toy model. A detection of forest lensing with this estimator therefore needs the complete DESI sample (roughly twice the DR1 forests and deeper coverage at $z > 2.5$) or DESI-II; DR1 can deliver the pipeline, the response measurement and upper limits. The response term is measured at only 2.7σ by the same three-dimensional pair statistic, well below the 4.8σ of Karaçaylı et al. [14], because their per-skewer $P_{1D} \times \kappa_{\text{CMB}}$ statistic is immune to the sparse-sampling noise that dominates here; the two measurements are complementary, and the P_{1D} bispectrum can be used to fix the response amplitude externally. Reaching 5σ requires ~ 50 available sightlines per deg^2 at fixed redshift with cleaned spectra, i.e. the complete DESI survey or DESI-II. Low-redshift galaxies as the lensing tracer avoid the deprojection but not the sampling noise, so they offer no gain for the same forest data; their value is as an independent check. Not included, and all in the direction of lowering these numbers: forest non-Gaussianity (a factor 1.5–2.7 in Shaw et al. 4), continuum-fitting distortions, the noisier high-redshift end of the slab, the uncertainty in $b_q(z)$ and m , and the use of a white CMB noise level instead of the released N_L . These are Gaussian toy forecasts; the estimator of Section 5 and the injection tests described there are what will turn them into a measured significance.

5 The pair-template estimator

The cell-based statistic of the original plan (now superseded) had two avoidable losses: a top-hat cell window that suppresses the response near $L \sim 1/\theta_{\text{cell}}$, exactly where the signal is, and the squeezed-limit harmonic decomposition, whose error relative to the exact response is a factor $\text{sinc}(\mathbf{L} \cdot \boldsymbol{\theta}_{ab}/2)$ per pair, equal to 0.80 at $L = 300$ for a $30 h^{-1} \text{ Mpc}$ pair aligned with $\mathbf{L}$ (check 7 in `mathematica/pair_template_checks.wls`). Both disappear if one uses the exact pair response of Eq. (5) with a filtered CMB map as the template. This section defines that estimator precisely; the definitions were sharpened after the second adversarial review (`report/reviews/codex_review.2.md`), whose corrections are noted where they apply.

5.1 Pair response

For a deflection field $\alpha(\theta)$ common to all pixels of a sightline (single source plane; relaxed below), the covariance of the pixel pair $i = (a, p)$, $j = (b, q)$ responds as

$$\delta C_{ij} = \nabla_\theta \xi(\theta_{ab}, \Delta\chi_{pq}) \cdot [\alpha(\theta_a) - \alpha(\theta_b)] + O(\alpha^2), \quad (28)$$

exactly at first order for any L and any sightline pattern, within the flat-sky homogeneous model in which ξ is the true covariance of the observed pixels. Here ξ is the mean cross-sightline correlation function *after* whatever line-of-sight operations (continuum projection, weights) the pipeline applies: for fixed projectors P_a , P_b acting along the two sightlines, $P_a(\nabla_\theta C_{ab})P_b^T = \nabla_\theta(P_a C_{ab} P_b^T)$ (check 5), so the transverse gradient of the observed pair covariance is what enters. Replacing the pair-specific projected covariance by the derivative of a single geometry-averaged measured $\xi(r_\perp, r_\parallel)$ is an approximation whose error is quantified on mocks (Section 5.5).

5.2 Amplitude estimator

Let X be an external map (the deprojected CMB convergence) and S_L a fiducial model for $C_L^{\kappa_{Ly\alpha} X}$. The quantity to be measured is the amplitude A in $C_L^{\kappa_{Ly\alpha} X} = A S_L$. Under a Gaussian conditional model, $\mathbb{E}[\kappa_{Ly\alpha}(\mathbf{L}) | X] = A h_L X(\mathbf{L})$ with the Wiener filter $h_L = S_L / C_L^{XX}$ (with C^{XX} the total, noise-included power of the map actually used). The template is therefore the deflection of the filtered map, $\alpha_T = \nabla \phi_T$, $\phi_T(\mathbf{L}) = 2h_L X(\mathbf{L})/L^2$, and for diagonal pixel weights w_i the estimator is

$$\hat{A} = \frac{1}{F} \left[\sum_{i < j} w_i w_j d_i d_j R_{ij} - b \right], \quad R_{ij} = \nabla_\theta \xi_{ij} \cdot \Delta \alpha_{T,ab}, \quad F = \sum_{i < j} w_i w_j R_{ij}^2, \quad b = \sum_{i < j} w_i w_j \xi_{ij} R_{ij}, \quad (29)$$

where the sums run over pixel pairs on distinct sightlines and $\Delta \alpha_{T,ab} = \alpha_T(\theta_a) - \alpha_T(\theta_b)$. Its properties, stated carefully:

- **What it measures.** Conditional on X , and to first order, $\mathbb{E}[\hat{A} | X] = (R_T, R_{\text{true}})_w / (R_T, R_T)_w$ with $(U, V)_w \equiv \sum_{i < j} w_i w_j U_{ij} V_{ij}$: the weighted projection of the true covariance perturbation onto the template response. It equals A when the true deflection conditional on X is $A \alpha_T$, which is what the Gaussian conditional model with $h_L = S_L / C_L^{XX}$ asserts. An earlier draft used $h_L \propto S_L / [(C_L^{\kappa_{Ly\alpha} \kappa_{Ly\alpha}} + N_\kappa(L)) C_L^{XX}]$; that filter is the right *weighting* for combining multipoles when N_κ varies with L , but it breaks the unit normalisation, and in the noise-dominated diagonal model the extra factor cancels against the unnormalised score response, so the template $h_L X$ is both correctly normalised and optimal there. The division by F , which depends on the realisation of X , makes $\hat{A}$ a conditional (template-coefficient) estimator rather than a fixed-weight cubic statistic; the cubic form is its numerator.
- **Bandpowers.** With B band-limited templates $\alpha_{T,b}$ the responses form a matrix $F_{bc} = \sum_{a < b} \Delta \alpha_{ab,b}^T M_{ab} \Delta \alpha_{ab,c}$ that is not diagonal on an irregular footprint; the band amplitudes are $\hat{\mathbf{A}} = \mathbf{F}^{-1}(\mathbf{q} - \mathbf{b})$ with q_b the band scores. $\mathbf{F}$ is computed from the pair catalogue (Section 5.3) at $O(B^2 N_{\text{pair}})$ cost and defines the band windows.
- **Mean field.** b is not small: $\xi \nabla \xi \cdot \Delta \alpha$ is even under pair reversal (both factors flip sign), so for a constant convergence it is $\propto -\kappa r \xi \xi'$ and survives orientation averaging. It is computed exactly for the actual geometry and template, which removes the modelled mean field given a correct ξ ; residual mean-field errors are tested on unlensed mocks with the real geometry.
- **Variance.** $1/F$ is the variance only for independent pair products. Pixel pairs sharing a sightline are correlated through $C_{aa} C_{bc}$ -type terms, and pairs (p, q) , (p', q') on the same two sightlines through the strong radial correlations $C_{aa}(p, p') C_{bb}(q, q')$; these correlations

can raise or lower the variance, so $1/F$ is a scale, not a bound. The covariance comes from jackknife resampling of the pair catalogue (regions assigned by pair midpoint) and from joint mocks, and the diagonal-weight estimator is suboptimal by an amount that includes correlations along each sightline, not only between neighbouring sightlines. Modes outside the science bands still couple to the band templates through the irregular geometry, so every fit includes a “junk” band covering the omitted multipoles.

- **Normalisation is model dependent through $\nabla\xi$,** exactly as any quadratic estimator is through its fiducial spectrum. $\xi(r_\perp, r_\parallel)$ at $r \leq 40 h^{-1}\text{Mpc}$ is measured from the same data with statistical precision far below the target accuracy on A ; the systematic question is whether the derivative of the measured, distorted, geometry-averaged ξ equals the pair-specific projected derivative, which is answered on mocks.
- **Redshift-dependent deflection.** The deflection of a pixel at χ by lenses at $\bar{\chi}_l$ scales as $(\chi - \bar{\chi}_l)/\chi$ relative to a reference plane, a variation of order 20 per cent across a $350 h^{-1}\text{Mpc}$ forest for $\bar{\chi}_l$ at $z \simeq 1$. With $g(\chi) \simeq 1 + g_1(\chi - \chi_{\text{ref}})$ the kernel ratio, the deflection difference of a pixel pair is $\Delta\alpha_{pq} = \Delta\alpha_{ab}(1 + g_1\Delta\chi_{\text{mid}}) + g_1\frac{1}{2}(\chi_p - \chi_q)(\alpha_a + \alpha_b)$ (check 12), so the catalogue stores, besides the zeroth moments, the first and second moments in $\Delta\chi_{\text{mid}} = \chi_{\text{mid}} - \chi_{\text{ref}}$ and the moments in $(\chi_p - \chi_q)$ that multiply the pair sum $\hat{\theta}_{ab} \cdot (\alpha_a + \alpha_b)$; the response F is quadratic in g , $m_0 + 2g_1m_1 + g_1^2m_2$ plus the cross terms, not linear (a linear prescription would return $A = 1.2$ for a unit signal with $g = 1.2$). This is the common-deflection approximation of Metcalf et al. [3] lifted to first order in the kernel variation.

5.3 Compression

The deflection depends only on the sightline pair, and $\nabla_\theta\xi$ is along $\hat{\theta}_{ab}$, so all pixel sums reduce to scalars per sightline pair,

$$v_{ab} = \sum_{p \in a, q \in b} w_p w_q d_p d_q \chi_{\text{mid}} \xi_{,r_\perp}, \quad m_{ab} = \sum_{p,q} w_p w_q (\chi_{\text{mid}} \xi_{,r_\perp})^2, \quad \beta_{ab} = \sum_{p,q} w_p w_q \xi \chi_{\text{mid}} \xi_{,r_\perp}, \quad (30)$$

with $\xi, \xi_{,r_\perp}$ evaluated at $(r_\perp, r_\parallel) = (\chi_{\text{mid}}\theta_{ab}, |\chi_p - \chi_q|)$, plus their first $(\chi_{\text{mid}} - \chi_{\text{ref}})$ moments and a small number of $(r_\perp, r_\parallel)$ bins for the shape test below. Then, with $d_{ab} = \hat{\theta}_{ab} \cdot \Delta\alpha_{T,ab}$,

$$\sum_{i < j} w_i w_j d_i d_j R_{ij} = \sum_{a < b} v_{ab} d_{ab}, \quad F = \sum_{a < b} m_{ab} d_{ab}^2, \quad b = \sum_{a < b} \beta_{ab} d_{ab}. \quad (31)$$

The compression is exact for this statistic with fixed kernel, weights and cuts; it does not preserve information for statistics not planned for (new radial bins, a different ξ model, different masks), which require a new pixel pass. The numerator can be compressed one level further: with $H_{ab} = (v_{ab} - \beta_{ab})\hat{\theta}_{ab}$ and $\mathbf{U}_a = \sum_{b>a} H_{ab} - \sum_{b<a} H_{ba}$, exactly $\sum_{a<b} H_{ab} \cdot \Delta\alpha_{ab} = \sum_a \mathbf{U}_a \cdot \alpha_a$: one vector per sightline gives the score for any template. (An earlier draft claimed no such compression exists; the local isotropic cancellation of check 3 concerns the *expectation* of $\mathbf{U}_a$ at first order, not the algebra.) The normalisation $F = \alpha^T \mathbf{L}_M \alpha$, a quadratic form with the block graph Laplacian built from $m_{ab}\hat{\theta}\hat{\theta}^T$, needs the pair level. For DESI DR1 there are $\sim 10^7$ sightline pairs within $30 h^{-1}\text{Mpc}$; the catalogue with eleven accumulators in six shape bins is ~ 2.6 GB in float32, and a template costs $O(N_{\text{pair}})$ for F and $O(N_q)$ for the score, exclusive of building the template map itself.

5.4 Templates and controls

- **Signal:** $X = \kappa_{\text{CMB}} - \hat{\kappa}_s$ with the kernel-matched quasar template (25), for ACT DR6 and Planck PR4 separately; $h_L = S_L / C_L^{XX}$ with $S_L = C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\text{CMB}}} - C_L^{\kappa_{\text{Ly}\alpha} \hat{\kappa}_s}$ from the fiducial model, so that the small lensing signal removed by the template is kept in S_L .

- **Response control:** $X = \hat{\kappa}_s$ measures the forest response to the slab density plus the (small, ~ 1 per cent) lensing correlation with the slab and its magnification part; it is a response measurement up to those terms, not a pure one.
- **Curl template:** $\alpha_R = R_{90}\alpha_T$. Its expectation vanishes for a parity-invariant ensemble and an isotropic pair distribution (check 6) but not for a given template on the real footprint: the gradient-to-curl response $F_{RT} = \sum m_{ab} d_{ab}^{(R)} d_{ab}^{(T)}$ is computed from the catalogue and the gradient and curl amplitudes are fitted jointly, so the curl amplitude is a null after the geometric leakage is accounted for. It is blind to parity-even contaminants.
- **Random-template ensemble:** independent CMB simulations κ_r filtered identically, each with its own b and F , give the distribution of $\hat{A}$ under no correlation between the forest and the CMB map. This is a conditional diagnostic of the CMB-noise and CMB-cosmic-variance part of the error; it is not a null for the deprojected statistic (the quasar subtraction is fixed) and it cannot detect a wrong mean field for the real template, since a zero-mean random template averages any mean-field error to zero.
- **Joint mocks:** the covariance of the deprojected amplitude, the mean field for the real template, and the residual response after deprojection are obtained from joint forest, quasar and convergence mocks with the real sightline geometry, selection and continuum operations (Section 5.5).

5.5 Validation, and what injections do and do not test

Shifting the observed sightline coordinates of the real data by $-A\alpha_{\text{inj}}$ and rebuilding the pair catalogue produces a data set whose pair covariances, expressed in the new coordinates, are those of a field lensed by $A\alpha_{\text{inj}}$ (sign: $\theta_{\text{obs}} = \theta_{\text{src}} - \alpha$). This tests the geometric bookkeeping of the pipeline: pair re-association, the sign and normalisation conventions of the templates, the band response matrix and the mean-field subtraction. It does *not* calibrate the physical response, because the kernel $\nabla\xi_{\text{mod}}$ is re-evaluated at the shifted positions and the recovered slope is the response of the *model* covariance, not of the true one: with a locally linear model $\xi_{\text{mod}} = c + gr$ the shifted score has slope g^2 whatever the data, whereas real lensing responds as $g\xi'_{\text{true}}$. The physical normalisation is fixed by $\nabla\xi$ measured from the data and validated on mocks in which the truth is known and the same measured- ξ procedure is applied: the mock must be generated with the sparse sightline geometry, the continuum projection, the noise distribution and, for the deprojection test, a forest response to long modes correlated with the mock's quasars and with the slab part of its κ_{CMB} . Paired $\pm A$ injections with the zero-injection result subtracted isolate the linear response from selection and boundary effects.

Two further limits of the tests: the second-order pair response $\frac{1}{2}\Delta\alpha_\mu\Delta\alpha_\nu\partial_\mu\partial_\nu\xi$ (check 1b) correlates with X only through $\langle\alpha\alpha X\rangle$, which vanishes for Gaussian fields but not in the presence of a lensing bispectrum; and none of the nulls above bounds parity-even contamination from shared sampling, magnification or residual response, which is why the joint mocks and the tomographic sub-slab tests are part of the first measurement rather than of a later phase.

5.6 Relation to previous estimators and cost

Equation (29) is the estimator of Metcalf et al. [3] with their many Legendre coefficients of ϕ replaced by a few template amplitudes, which removes the ill-conditioned Fisher inversion that limited their maps, and it is the continuum estimator of Section 2.6 with the exact discrete geometry and C^{-1} approximated by inverse-variance pixel weights. Costs: one pixel-pair pass over $\sim 1.4 \times 10^{11}$ unique pairs, to be benchmarked (the arithmetic bound is minutes on 24 threads; realistic kernels with interpolation and scattered access are 10–50 times slower); a new pass for every injection, mock, change of ξ model, weights, masks or unstored bins; $O(N_{\text{pair}})$ per template

and $O(B^2 N_{\text{pair}})$ per band-response matrix afterwards, plus the map-level cost of building the templates.

6 Caveats and next steps

- **Single source plane.** Pixels at different χ are lensed by slightly different convergences; the estimator as written recovers a $k_{\parallel}$ -weighted average. Modelling this is straightforward within (9) (each screen gets its own W_s).
- **Gaussianity.** The forest is strongly non-Gaussian on the scales that carry the information. Shaw et al. [4] find a factor 1.5–2.7 loss in signal-to-noise; the variance (12) will need the forest trispectrum or simulations.
- **Clustering bias.** The dominant systematic is the intrinsic $\langle \delta_F \delta_F \kappa_{\text{CMB}} \rangle$ bispectrum. The plan is to compute its projection onto $\hat{\kappa}$ using the measured responses of Chiang et al. [16] and the DESI Y1 measurement of Karaçaylı et al. [14], and to design a hardened estimator that is orthogonal to both the density and tidal responses; the price is the factor ~ 8 in noise found above for the density part alone.
- **Data-level effects.** Continuum fitting removes low- $k_{\parallel}$ modes and distorts the measured ξ ; DLAs, metals and quasar magnification bias [17] add non-lensing structure; the CMB maps carry tSZ/CIB residuals correlated with the forest’s foregrounds.
- **Numerics to firm up.** Replace the white CMB noise by the public ACT DR6 and Planck PR4 noise curves; refine the Arinyo parameters to the published $z = 2.4$ values; add pixel noise and a realistic $n(z)$ of forests.
- **Pending adversarial review.** The derivations in Sections 2.2–2.6 and the numbers in Section 3 are queued for review with codex/gpt-6-astra, which was unavailable when they were written.

A Symbolic checks

The scripts in `mathematica/` verify: (A) the second-order squeezed expansion (6), its angular average $2\mathcal{P} + l\mathcal{P}'$, its $\cos 2\varphi$ component $l\mathcal{P}'$, and the power-law result $(2+n)\mathcal{P}$; (B) that the exact power spectrum of a field dilated and sheared by constant $\kappa, \gamma_1, \gamma_2$ agrees with (6) to first order, and that $\kappa = -\nabla^2 \phi / 2$ for the potential used; (C) that the real-space pair-displacement covariance (5), Fourier transformed in a periodic one-dimensional toy model, reproduces the mode coupling (4) for all tested (L, l) and vanishes for $l + l' \neq L$. `code/recon_noise.py` contains the white-spectrum mode-counting check of (15).

B Response to the first adversarial review

The codex review of 2026-09-10 (`report/reviews/codex_review_1.md`) confirmed the continuum normalisation and mode counting of Section 2.6 and raised 18 items. Changes made in response: the potential-to-convergence conversion and the mask mixing caveat in Section 2.5 (item 1); the removal of the incorrect $N^{(1)}$ -type argument (item 2); the factor 1/2 between the power response R_δ and the field modulation, which halves the contamination estimates (item 3); the kernel-matching condition (24) and the matched template (25), replacing the single slab-averaged coefficient (item 4); the consistent magnification treatment and the first-order nature of coefficient errors (items 5, 6); the withdrawal of the “quasars behind the slab” remedy in favour of tomographic sub-slabs (item 7); the corrected deprojected variance including the response sample variance (item 8); the cutoff-sensitivity and mode-coupling caveats of the noise model (item 9);

the sightline density $\bar{n}_{\text{los}} = 22$ and the distinction from a weighted effective density (item 10); the Limber integral for $C^{\kappa_{\text{CMB}}\kappa_{\text{CMB}}}$ extended to recombination with the CMB noise recalibrated (item 11); the $40 \leq L \leq 300$ headline range (item 12); the replacement of the cell-based estimator by the exact pair-template estimator of Section 5 (items 13, 15); injection calibration of the response through the continuum projection (item 14); and the revised validation plan in `PLAN.md` (items 16–18). Not yet done: the higher-order lensing corrections, a joint likelihood forecast with $b_q(z)$ and m free, validation of the sparse-sampling noise model against the discrete estimator, and the benchmark of the pair kernel.

A third, confirmation review (`report/reviews/codex_review_3.md`) found two errors in the implementation specification that the report text had not caught: the tangent-plane separation was written with the opposite orientation to $\theta_{ab} = \theta_a - \theta_b$, which reverses the sign of the recovered amplitude (check 11), and the redshift-dependent deflection had been treated linearly where F is quadratic in the kernel ratio and a $(\chi_p - \chi_q)$ cross term appears (check 12). Both are corrected above and in `IMPLEMENTATION.md`, together with: the completeness division and finite-random noise of the matched map; a mock in which the forest lensing includes the lenses inside the slab, quasars trace the unsmoothed density with redshift-space distortions and magnification, and the real mask can be applied; a junk band for omitted multipoles; the $1/F$ wording; disjoint template and sightline selections per sub-slab; and a set of eleven acceptance tests with predeclared tolerances that separately probe the physical normalisation (fixed-geometry remapping at several amplitudes), the mean field (unlensed mocks with fixed templates), the stochastic normalisation, a response-only null and the combined recovery.

The second review (`report/reviews/codex_review_2.md`), restricted to the estimator, confirmed the exactness of the pair response and of the pair-level compression and raised 16 items, all folded into Section 5 and `IMPLEMENTATION.md`: the amplitude is now defined as a conditional template coefficient with the filter $h_L = S_L/C_L^{XX}$ (item 1); band amplitudes use the full response matrix (2); the mean-field oddness claim was withdrawn (3); injection tests are classified as pipeline tests, with the physical normalisation fixed by the measured $\nabla\xi$ and validated on mocks (4, 5); the per-sightline numerator compression is stated and the retained bins specified (6); $1/F$ is a lower bound on the variance (7); the curl amplitude is fitted jointly with its geometric leakage (8); the random-template ensemble and the joint-mock covariance are distinguished (9); the matched template gets redshift margins, an RSD tolerance and the general shot-noise formula (10, 11); the magnification and stochasticity statements were corrected (12, 13); shared sampling and tomography are in the first measurement (14); the second-order term is no longer called an $N^{(1)}$ bias (15); and cost statements separate contractions from map building and catalogue rebuilding (16).

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